

EXP.NO: 4	AWK Scripting
DATE: 30/01/2026	

AIM:

To study and implement AWK scripting for processing text files, performing mathematical calculations, and generating formatted outputs based on specific conditions.

ALGORITHM:

1. Initialization: Prepare an input data file named data.txt containing structured text with multiple fields (e.g., ID, Name, Salary, and Scores).
2. Input Reading: Read the input file line by line using the AWK utility.
3. Field Extraction: Use positional parameters (\$1, \$2, \$3, etc.) to access specific data columns and built-in variables like NR for row numbers and NF for the count of fields.
4. Summation Logic:
 - Initialize a accumulator variable (e.g., total = 0) in the BEGIN block.
 - Add the value of the target field (Salary) to the accumulator for every record processed.
 - Display the final sum in the END block after all lines are read.
5. Conditional Filtering: Apply if statements to check specific criteria, such as identifying employees with a salary greater than a defined threshold (e.g., 6000).
6. Complex Logic Implementation:
 - Evaluate multiple fields simultaneously using logical operators (e.g., || for OR).
 - Check if any subject score is below the passing mark (e.g., 45).
 - Assign and print a status of "Pass" or "Fail" based on the result of the logical evaluation.
7. Output Generation: Format and print the processed results to the standard output (terminal).

CODE:

```
101 Ravi 5000 5 60 55 70
102 Sita 7000 6 40 50 65
103 Arun 6500 3 80 85 90
104 Meena 8000 7 30 60 75
105 Kiran 5500 4 72 68 74
```

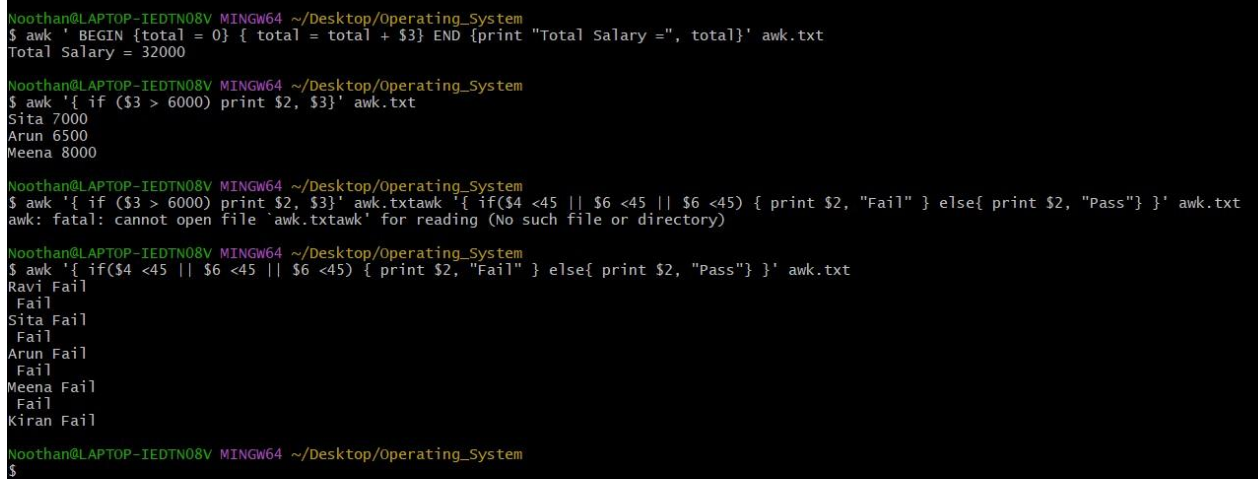
```
awk '
BEGIN {total = 0}
{
    total = total + $3
}
END {
```

```
    print "Total Salary =", total
}' data.txt
```

```
awk '{    if ($3
> 6000)
print $2, $3
}' data.txt
```

```
awk '{    if ($5 < 45 || $6 < 45 ||
$7 < 45)
    print $2, "Fail"
else    print $2,
"Pass" }' data.txt
```

OUTPUT:



The screenshot shows a terminal window with the following commands and output:

```
Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System
$ awk ' BEGIN {total = 0} { total = total + $3} END {print "Total Salary =", total}' awk.txt
Total Salary = 32000

Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System
$ awk '{ if ($3 > 6000) print $2, $3}' awk.txt
Sita 7000
Arun 6500
Meena 8000

Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System
$ awk '{ if ($3 > 6000) print $2, $3}' awk.txtawk '{ if ($4 < 45 || $6 < 45 || $6 < 45) { print $2, "Fail" } else{ print $2, "Pass"}}' awk.txt
awk: fatal: cannot open file `awk.txtawk' for reading (No such file or directory)

Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System
$ awk '{ if ($4 < 45 || $6 < 45 || $6 < 45) { print $2, "Fail" } else{ print $2, "Pass"}}' awk.txt
Ravi Fail
Fail
Sita Fail
Fail
Arun Fail
Fail
Meena Fail
Fail
Kiran Fail

Noothan@LAPTOP-IEDTN08V MINGW64 ~/Desktop/Operating_System
$
```

RESULT:

Thus, AWK scripting was successfully implemented to process text data, perform arithmetic calculations, and apply logical conditions to generate meaningful and formatted output.